Handheld SOP

Station 3

How to select the master batch source pallet for Station 3 with one scan, read what Station 3 sends back, and what to do when it refuses a scan.

Version	Date	Device	Station
1.5.0	2026-11-06	Chainway C72	Station 3 master batch

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01 Purpose and scope

The Station 3 app runs on the Chainway C72 handheld at the master batch station. Operators use it for one job: **selecting the source pallet** that master batch will be drawn from. You scan the pallet, and Station 3 selects it for you and shows its product and how many kilograms are available. You then make the packs on the Station 3 desktop.

What it does

- Sends the pallet barcode or RFID tag you scan to Station 3, which selects that pallet as your source.
- Shows the source's product code and description, the kilograms available, and the pallet and batch.
- Tells you when a pallet cannot be used, and why.

What it does not do

- It does not weigh, make packs or print labels. You enter the pack weight on the Station 3 desktop, signed in as the same user. Each new label deducts its weight there.
- Scanning never deducts stock. It only chooses the source.
- It keeps nothing to send later. Every scan needs Station 3 online.

> **Who this is for**
>
> Master batch operators at Station 3. Sections marked **Supervisor** are for the shift supervisor, who holds the settings PIN.

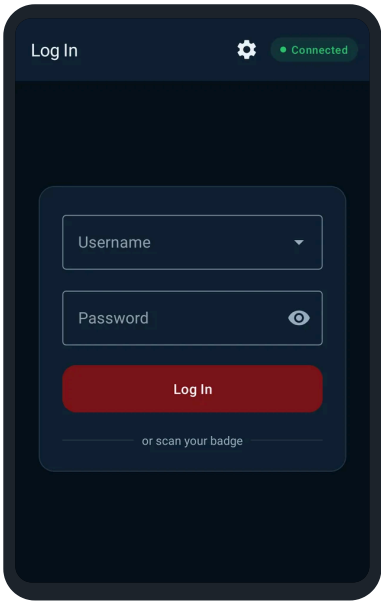
02 Before you start

- [ ] The handheld is charged. Put it back on its charger at the end of the shift.
- [ ] You are on the plant Wi-Fi, in coverage at the master batch station.
- [ ] You know your Station 3 username and password, or you have your RFID badge.
- [ ] The Station 3 desktop is running. The status pill shows **Connected** (section 3).
- [ ] The source pallet has its barcode label or its current RFID tag.

Opening the app

The handheld starts in the **PPNAM Station Launcher**. Tap the crimson **Station 3** tile.

The app opens on **Log In**, or straight on the Master Batch screen if you are still signed in. Wait a few seconds for the status pill in the top-right corner to settle on **Connected** before you sign in.



The Log In screen, ready for a username or a badge.

03 The status pill and header

The coloured pill in the top-right corner is on every screen. It reports two links: the **network link** (the broker that carries messages) and **Station 3 itself** (the desktop). It waits 1.5 seconds before it changes, so a short blip does not make it flicker.

● Connected

The network is up and the Station 3 desktop is running. Everything works.

What to do: Carry on.

● Reconnecting

The handheld lost the network link and is getting it back. It retries by itself, from every second up to every 30 seconds.

What to do: Wait. Scanning is paused until it clears. If it does not clear, move into Wi-Fi coverage.

● Station 3 offline

The network is fine, but the Station 3 desktop is not running or not reachable. A full-screen warning covers the Master Batch screen.

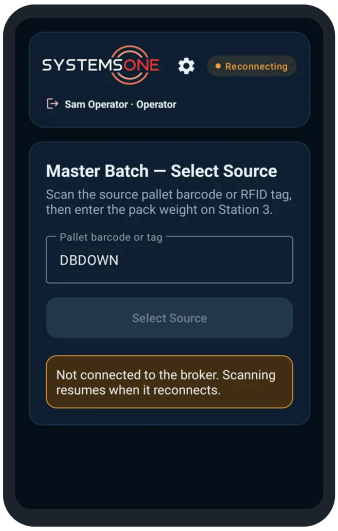
What to do: Wait. It clears by itself when Station 3 is back. Tell your supervisor if it lasts.

● Offline

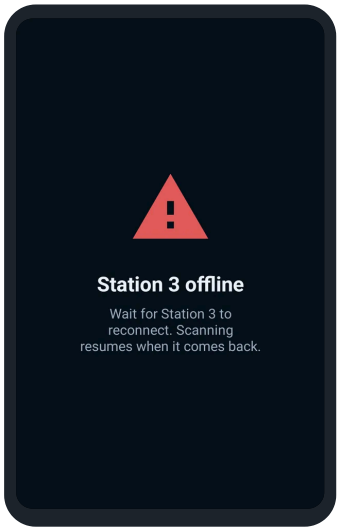
The handheld is not trying to connect: it has no broker account set up yet, or a supervisor is applying new settings.

What to do: Ask your supervisor to set up the broker in Settings.

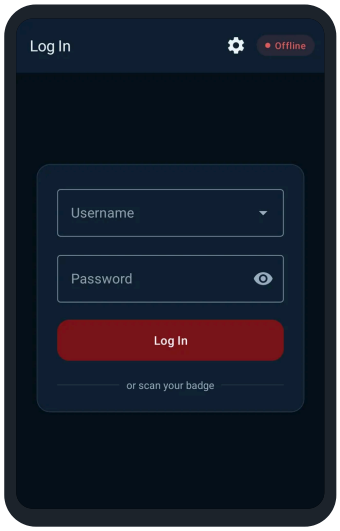
A red **Broker login rejected** pill means the broker refused this handheld's broker username or password. The app stops retrying until a supervisor corrects the broker account in Settings.



Reconnecting: scanning pauses until the link is back.



Station 3 offline: the screen is covered until it returns.

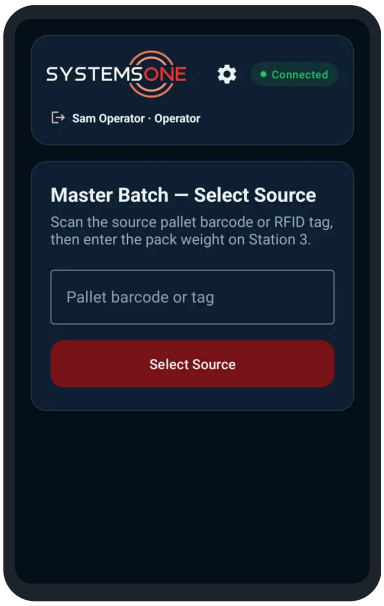


Offline: no broker account set up on this handheld.

The header

- **Your name and role** sit under the logo. Tap them to log out (section 10).
- The **gear** opens Settings.

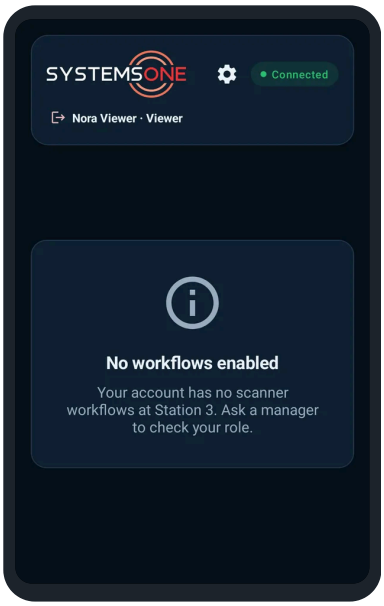
Below the header is the only job, **Master Batch — Select Source**. It is the home screen.



The Master Batch screen, ready for a scan.

No workflows enabled

Station 3 decides what each account may do. If yours may not select sources, the home screen shows *No workflows enabled* instead, and scans do nothing. Ask a manager to check your role on the Station 3 desktop.



This account may not select sources.

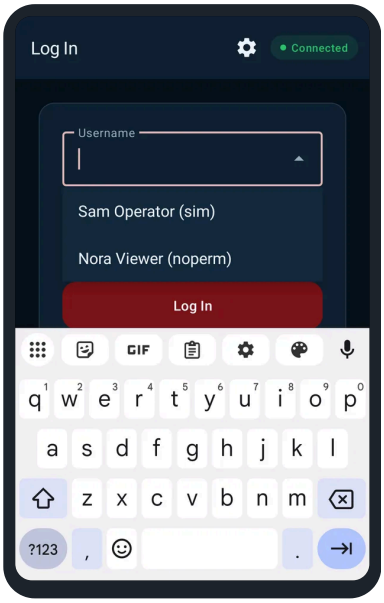
04 Signing in

Your source selection is recorded against your name, and the desktop expects the same user to make the packs, so the app does nothing until you sign in. You can sign in with your username and password, or with your badge.

By username and password

1. Tap **Username**. Pick your name from the list (each row reads *Your Name (username)*) or type your username.
2. Tap **Password** and type your password. The eye icon shows what you typed.
3. Tap **Log In**, or press Enter on the keyboard.

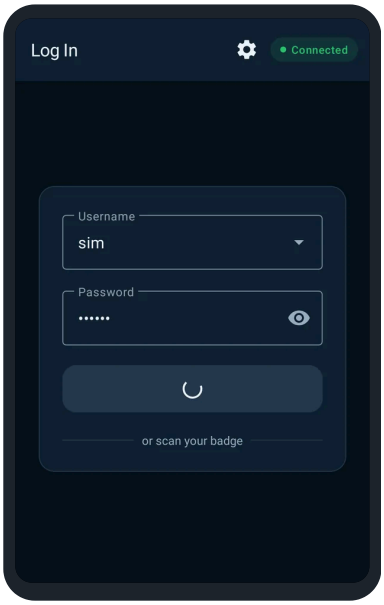
The list comes from Station 3. A username that is not in the list still works.



Tap the arrow to pick your name from the station's list.

While Station 3 checks

A spinner replaces the button label and both fields are locked until Station 3 answers, for up to 10 seconds. Your password itself is never sent over the network.

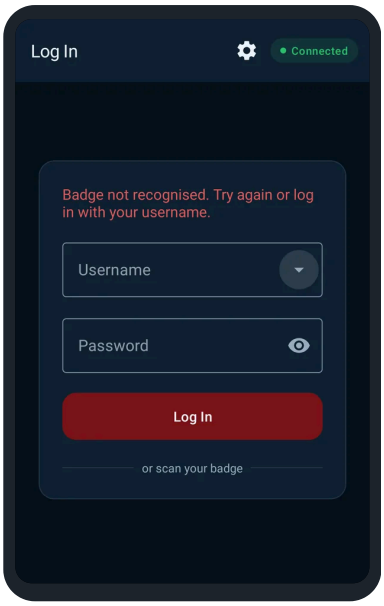


The spinner shows while Station 3 checks your password.

By badge

While the Log In screen is showing, **scan your RFID badge**. You do not need to tap anything first.

A second badge read while a sign-in is already running is ignored, so a double read cannot sign in the wrong person. An unknown badge gives *Badge not recognised. Try again or log in with your username.* Signing in makes no scan sound.



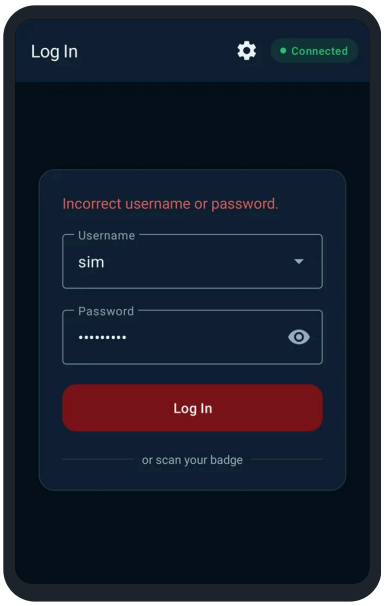
A badge Station 3 does not know is refused.

If sign-in fails

The reason appears in red at the top of the card. The common ones:

- **Incorrect username or password.** Retype them carefully.
- **Station 3 did not respond. Check the station and retry.** No answer within 10 seconds. Check the pill, then tap **Log In** again.
- **Not connected to the broker. Check Settings and retry.** The handheld has no network link.
- **Please fill in all fields** Both boxes are needed.

Every sign-in message is listed in section 8.

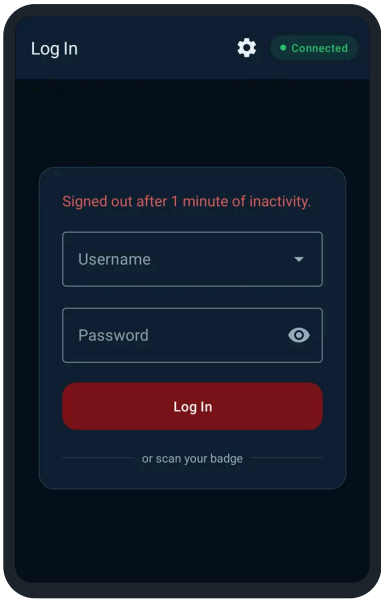


The reason appears in red above the fields.

Why you were signed out

- If nobody touches the handheld or scans for **15 minutes**, the app signs you out with *Signed out after 15 minutes of inactivity.* Your supervisor can change the time. Time with the app in the background counts too.
- The app checks your session with Station 3 every 30 seconds. If Station 3 has ended it, you return to Log In with Station 3's reason, usually *Sign in again.*
- When your session reaches the end time Station 3 gave it, you see *Your session expired. Sign in again.*

Station 3 going offline does **not** sign you out: the screen is covered until it returns.



Why you were signed out is shown on Log In.

05 Selecting a source

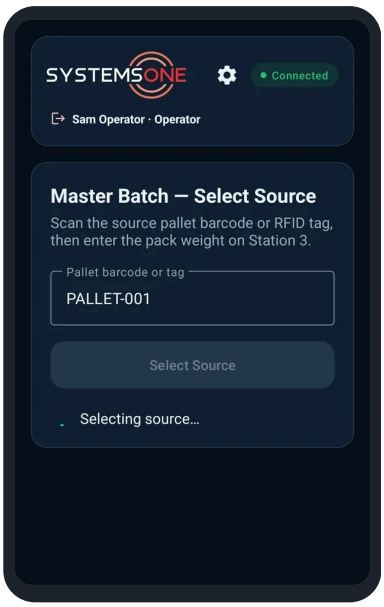
One scan selects one source. A new scan replaces the previous selection on Station 3.

Step 1 Scan the source pallet

Scan the source pallet's **barcode** or its **current RFID tag**. The scan is sent straight away: you do not need to tap anything.

You can also type the barcode into **Pallet barcode or tag** and tap **Select Source**, or press Enter.

Selecting source... shows while Station 3 looks the pallet up. **Select Source** is grey until it answers, for up to 10 seconds.



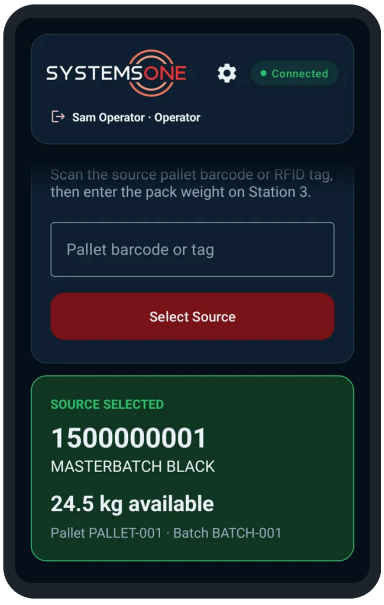
Selecting source... until Station 3 answers.

Step 2 Read the source

A green **Source selected** card appears and you hear the success chirp. It shows:

- the **product code** in large figures, with the product description under it;
- the **kilograms available** on that pallet;
- the **pallet** and **batch**.

Check it is the source you meant. The card clears itself after 15 seconds, and the field is empty for the next scan.

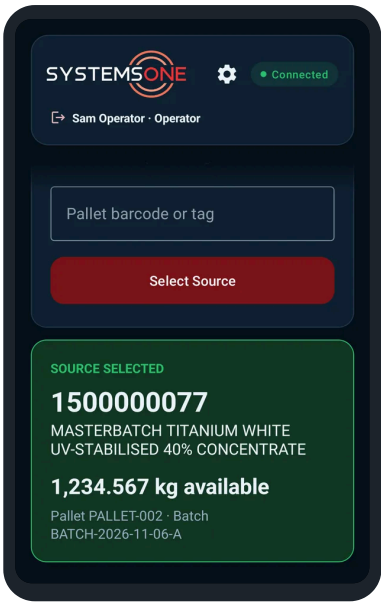


Selected. The kilograms available are what you can draw on.

Step 3 Make the packs on the desktop

Go to the Station 3 desktop, signed in as the **same user**, and enter the pack weight in kg there. Each new label deducts its weight from the source.

To change source, scan the other pallet. Weights show up to three decimals with thousands separators, for example `1,234.567 kg`.



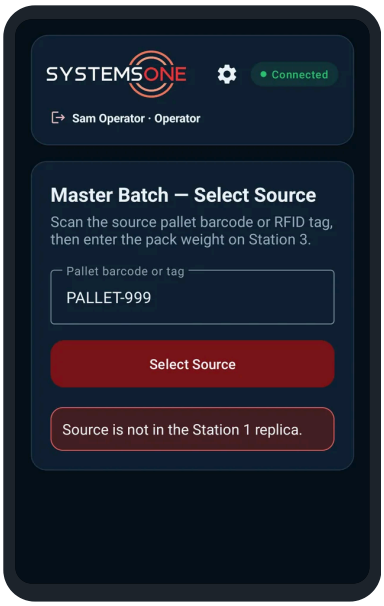
A long product description wraps; nothing is cut off.

If the source is refused

A red panel shows Station 3's reason and you hear the fail buzz. No source is selected. The usual ones:

- *Source is not in the Station 1 replica.* Station 3 does not know this pallet. Check you scanned the pallet's own label or tag.
- *Source stock is empty or unavailable. Wait for Central stock and source identity to sync.* The pallet has no stock Station 3 can use yet. Wait, then scan again, or choose another pallet.
- *This scan matches more than one source pallet. Call a supervisor.*

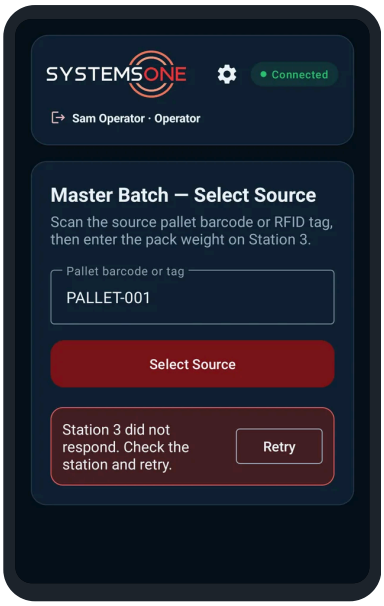
Section 8 lists every message.



A refused source. Scan a different pallet.

If Station 3 cannot answer

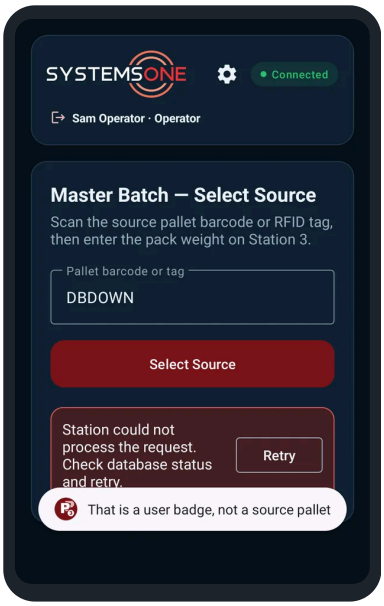
Station 3 did not respond. Check the station and retry. or *Station could not process the request. Check database status and retry.* comes with a **Retry** button. Check the pill, then tap **Retry**: it sends the last scan again. Scanning the pallet again does the same.



No answer in 10 seconds. Retry sends the same scan again.

A badge instead of a pallet

If you scan a staff badge, the app refuses it on the handheld with *That is a user badge, not a source pallet* and the fail buzz. Nothing is sent.



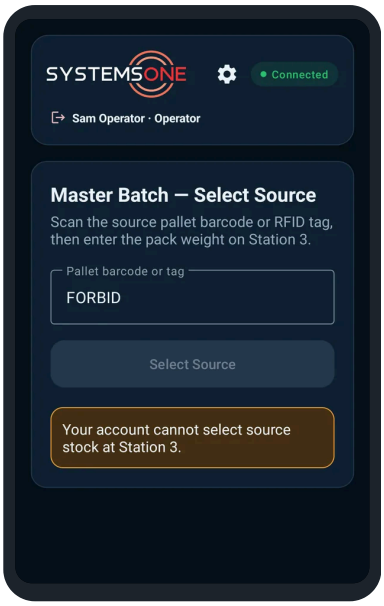
Badges are never sent as a source.

When scanning is paused

An amber panel means scanning is paused, and **Select Source** is grey:

- *Not connected to the broker. Scanning resumes when it reconnects.* Wait for **Connected**.
- *Station 3 is offline. Scanning resumes when it comes back.* Wait for **Connected**.
- *Your account cannot select source stock at Station 3.* Station 3 refused your account. Ask a manager to check your role, then sign in again.

A scan while paused is refused with the fail buzz and is not sent.



An amber notice says why scanning is paused.

06 Scan sounds and vibration

Every PPNAM handheld uses the same two sounds. The Chainway scanner beeps by itself when it reads anything; the app's sound comes **after** that, when the answer is known.

- **Success chirp** (a bright rising two-note chirp) with one short 40 ms vibration: the scan was **accepted**.
- **Fail buzz** (a low falling double buzz) with two 200 ms vibrations: the scan **failed**. Read the screen.

The sounds play at the **media volume**. If you cannot hear them, turn the media volume up; the vibration still tells you.

What you hear in the Station 3 app

When	Outcome	Sound
Master Batch	Station 3 selects the source	● **Success chirp**
Master Batch	Source refused, account not allowed, or Station 3 could not process it	● **Fail buzz**
Master Batch	No answer in 10 seconds, or the scan could not be sent	● **Fail buzz**
Master Batch	A staff badge read instead of a pallet	● **Fail buzz**
Master Batch	A scan while scanning is paused (offline, not allowed, or still waiting)	● **Fail buzz**
Master Batch	A typed value that is empty, too long or has hidden characters	● **Fail buzz**
Master Batch	Another read of the pallet already being looked up	○ **Silent**
Log In, Settings, buttons, dialogs	Signing in, tapping, the 30-second session check	○ **Silent**

The repeat-read rule

One pull of the trigger can read the same code several times. A repeat of a value that was **just refused on the handheld**, within 1.5 seconds, makes no sound, so holding the trigger never buzzes twice. A different value plays its own sound. A repeat read of the pallet Station 3 is already looking up is the same scan: it sounds once, on Station 3's answer.

07 Messages and what to do

Every message the app shows, word for word. Station 3 writes its own refusal reasons; the ones it uses today are listed by where they appear. *Ask a manager* or *Ask a supervisor* means the problem cannot be fixed on the handheld.

Signing in

Message	What it means	What to do
Please fill in all fields	The username or password box is empty.	Fill in both.
Incorrect username or password.	Station 3 did not accept the password.	Retype carefully. Ask your supervisor if you are locked out.
Badge not recognised. Try again or log in with your username.	The badge is not registered at Station 3.	Scan again, or sign in with your username and ask a manager to register the badge.
Login was refused by the station (*code*). Ask a supervisor.	Station 3 refused for another reason.	Tell your supervisor the code.
Station 3 could not complete the login. Try again.	Station 3's answer could not be used.	Try again. Report it if it repeats.
Not connected to the broker. Check Settings and retry.	The handheld has no network link.	Check the pill and Wi-Fi, then retry.
Station 3 did not respond. Check the station and retry.	No answer within 10 seconds.	Check that the Station 3 desktop is running, then retry.
Signed out after *N* minutes of inactivity.	The automatic sign-out ran.	Sign in again.
Sign in again. / Session ended. Sign in again.	Station 3 ended your session.	Sign in again and scan again.
Your session expired. Sign in again.	The session reached its end time.	Sign in again.

Selecting a source

Message	What it means	What to do
Selecting source…	Waiting for Station 3.	Wait.
Source selected	Station 3 selected the pallet.	Make the packs on the desktop.
That is a user badge, not a source pallet	A staff badge was read.	Scan the pallet.
Scan a barcode or tag of up to 500 characters.	The typed value is empty, too long or has hidden characters.	Scan the label, or retype it.
Still waiting for the previous scan.	A scan is already being looked up.	Wait for the answer.
Source is not in the Station 1 replica.	Station 3 does not know this pallet.	Check the label or tag. Ask a supervisor if it is the right pallet.

Message	What it means	What to do
Source stock is empty or unavailable. Wait for Central stock and source identity to sync.	The pallet has no usable stock yet.	Wait and scan again, or use another pallet.
Source stock changed or is unavailable. Scan again after checking the replica.	The stock changed during the scan.	Scan again.
This scan matches more than one source pallet. Call a supervisor.	Two pallets share this code.	Call a supervisor.
Scan a valid source barcode or tag. / Scan a source barcode or current RFID tag.	Station 3 could not use the value.	Scan the pallet's label or current tag.
Station 3 did not respond. Check the station and retry.	No answer within 10 seconds.	Check the pill, then **Retry**.
Station could not process the request. Check database status and retry.	The Station 3 database had a problem.	**Retry**. Tell your supervisor if it repeats.
Not connected to the broker	The scan could not be sent.	Wait for **Connected**, then **Retry**.
Could not reach Station 3	The scan could not be sent.	Check the pill, then **Retry**.
Station 3 sent an incomplete source	Station 3's answer was missing values.	**Retry**. Report it if it repeats.
Sign in before scanning	No one is signed in.	Sign in.
Not connected to the broker. Scanning resumes when it reconnects.	The network link is down.	Wait for **Connected**.
Station 3 is offline. Scanning resumes when it comes back.	The Station 3 desktop is not running.	Wait for **Connected**.
Your account cannot select source stock at Station 3.	Station 3 does not allow your account to select sources.	Ask a manager to check your role, then sign in again.
Station 3 offline / Wait for Station 3 to reconnect. Scanning resumes when it comes back.	The full-screen warning while Station 3 is offline.	Wait. It clears by itself.
No workflows enabled / Your account has no scanner workflows at Station 3. Ask a manager to check your role.	Your account has no Station 3 jobs.	Ask a manager.

Settings

Message	What it means	What to do
Enter the supervisor PIN.	The PIN box is empty.	Supervisor only.
Incorrect PIN. *N* attempts left before lockout.	Wrong PIN.	Supervisor only.
Too many attempts. Try again in *N*s.	Five wrong PINs lock the box for 30 seconds.	Wait for the countdown.

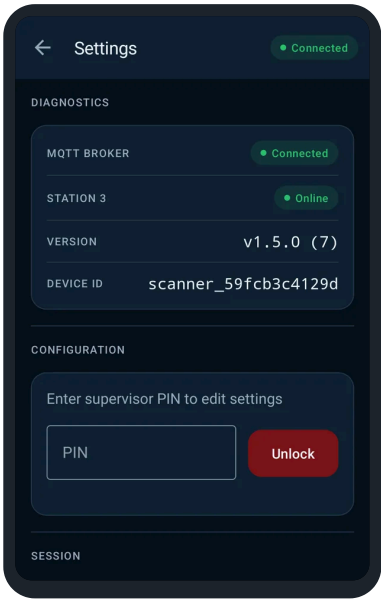
Message	What it means	What to do
Enter the broker host / Enter a port from 1 to 65535 / Enter 0–1440	A field is empty or out of range.	Supervisor: correct the field.
Could not store the password securely / Nothing was changed: the password could not be stored securely.	The handheld could not save the password.	Retry; report it if it repeats.
Saved, but could not connect to *host*:*port*. Check the broker settings.	The new settings were saved but do not connect.	Supervisor: check the values.
The broker rejected the username or password. Settings were saved — check them and try again.	Wrong broker account.	Supervisor: correct the broker username or password.

08 Settings

The gear opens Settings from the Log In and Master Batch screens. The top half is open to everyone and is what support will ask you to read out. The bottom half needs the supervisor PIN.

Diagnostics: anyone

- **MQTT Broker**: the network link on its own (Connected, Reconnecting, Offline).
- **Station 3**: the desktop (Online, Offline, Unknown). Broker *Connected* with Station 3 *Offline* means the network is fine and the station is the problem.
- **Version**: `v1.5.0 (7)` for this release.
- **Device ID**: this handheld's identity, such as `scanner_59fcb3c4129d`. Read it out when you report a fault.
- **Session** at the bottom shows who is signed in, with **Log out**.



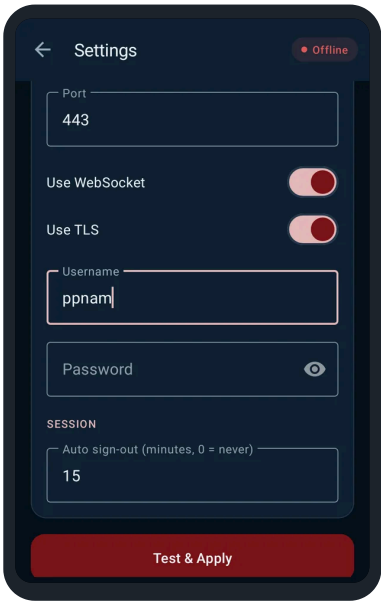
Diagnostics, and the PIN gate below.

> **Configuration**
> Type the supervisor PIN and tap **Unlock**. Five wrong attempts lock the PIN box for 30 seconds. Never tell an operator the PIN; operators should ask the supervisor. A supervisor's Settings RFID tag also opens Settings from any Station 3 screen; the PIN is still needed.

Unlocked: supervisors only

- **Host** and **Port**: normally `mqtt.sysone.co.za` and `443`.
- **Use WebSocket** and **Use TLS**: both on for the standard setup.
- **Username** and **Password**: the broker account, not an operator account. The saved password is never shown; leave the box blank to keep it.
- **Auto sign-out (minutes, 0 = never)**: 15 by default, up to 1440.

A new handheld has no broker password and stays **Offline** until one is saved here once.

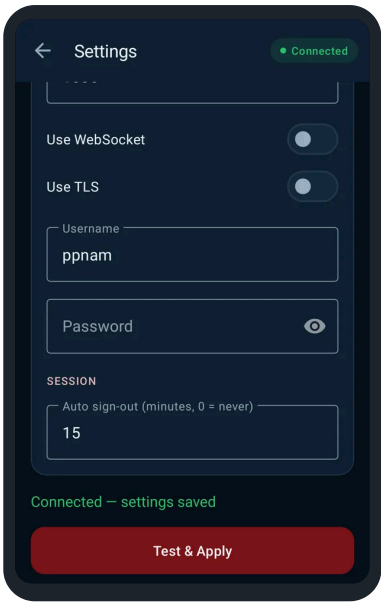


Unlocked, showing the standard broker settings.

Test & Apply

Test & Apply saves the settings, then reconnects with them while *Testing connection…* shows. The operator stays signed in.

- *Connected — settings saved*: done.
- *Saved, but could not connect to host:port. Check the broker settings.* or *The broker rejected the username or password. Settings were saved — check them and try again.*: correct the values and tap **Test & Apply** again.



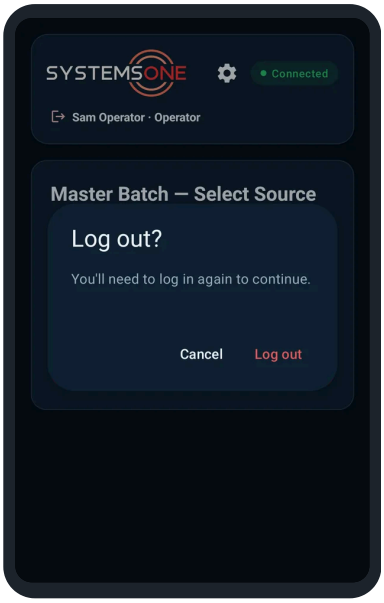
Connected — settings saved.

09 Finishing up

Handing the handheld over

1. Finish the packs for your source on the desktop.
2. Tap your name in the header (or **Log out** in Settings).
3. Confirm **Log out**.

Always log out before you pass the handheld to someone else. Otherwise their source selection is recorded under your name.

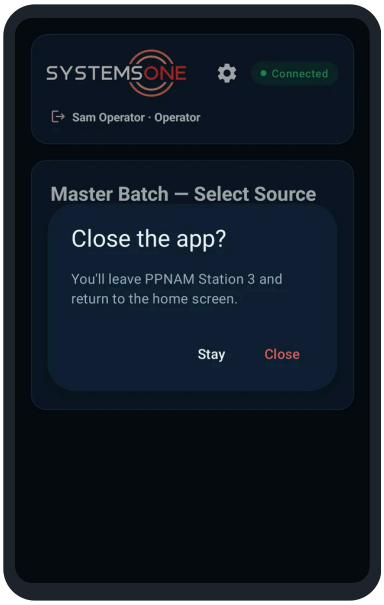


Log out asks first.

Closing the app

From the Log In or Master Batch screen, the **Back** key asks *Close the app?*. Tap **Close** to return to the Launcher, or **Stay**.

At the end of the shift, log out and put the handheld on its charger.



Back from Log In or the Master Batch screen.

10 Troubleshooting

Common problems

Problem	Likely cause	What to do
The screen is covered by Station 3 offline.	The Station 3 desktop is off, or its program is not running.	Wait. Tell your supervisor if it lasts.
The pill stays Reconnecting.	No Wi-Fi where you are.	Move into coverage. If it stays, your supervisor checks Settings.
The pill is red Offline on a new handheld.	No broker account has been saved.	Your supervisor sets it up in Settings.
The pill is red Broker login rejected.	The broker account in Settings is wrong.	Your supervisor corrects it in Settings.
***No workflows enabled* after sign-in.**	Your account has no Station 3 jobs.	Ask a manager.
Select Source stays grey.	Scanning is paused: read the amber notice.	Wait for **Connected**, or ask a manager about your role.
Every scan says *Source is not in the Station 1 replica.*	Wrong label, or the pallet has not reached Station 3's copy of Station 1 stock.	Check the label. Tell your supervisor.
The source card disappeared.	It clears itself after 15 seconds.	Scan the pallet again to see it.
No scan sounds.	Media volume is down.	Turn up the media volume. The vibration still works.
Signed out unexpectedly.	15 minutes without a touch or scan, or Station 3 ended the session.	Read the message on Log In and sign in again.

Reporting a fault

Give your supervisor the **Device ID** and **Version** from Settings, the time, the pallet barcode, and the exact message on screen.

11 Revision history

Version	Date	Change
`1.5.0`	2026-11-06	Suite release: visual refresh, consistency pass, scan sounds. First Station 3 handheld SOP.